

PHOTOGRAPHIC PHYSICS & IMPERFECTIONS

The Complete Technical Glossary of Real-World Camera Phenomena, Optical Physics, and Authentic Photographic Imperfections

Version 1.1 · April 2025 — Corrected Edition

For use with: AuthenticLens — The Realism Architect Gemini Gem
Confidential Reference Document

This document is the definitive photographic physics and optics reference for the AuthenticLens Gemini Gem. It provides technically accurate definitions of every camera phenomenon, optical imperfection, and real-world photographic artifact that distinguishes genuine photography from AI-generated imagery. These concepts form the vocabulary and theoretical foundation for Phase 2 realism enhancement protocols.

SECTION 1 — OPTICAL ABERRATIONS

Optical aberrations are systematic deviations from perfect image formation that occur in all real lenses. Unlike AI-generated images, which can produce geometrically perfect images, real photographs always contain measurable aberrations. Their presence and character are determined by the specific lens design, aperture setting, and focal length.

1.1 — CHROMATIC ABERRATION (CA)

■ ERRATA — Doc 2 § 1.1 CORRECTION

ORIGINAL TEXT stated: 'AI images show neither CA nor correction artifacts.' This is REVISED. Modern generative models can simulate optical flaws, though often inconsistently. Absence, over-regularity, or inconsistent simulation of aberrations can be informative, but absence alone is not decisive. Use CA behaviour as a contextual cue, not primary evidence.

Chromatic aberration occurs because optical glass has a different refractive index for different wavelengths. Red, green, and blue light are bent by different amounts as they pass through a lens, causing them to focus at different distances, producing colour fringing around high-contrast edges.

Two Primary Types:

- LONGITUDINAL (AXIAL) CA: All wavelengths fail to converge at the same focal plane. Manifests as magenta/purple fringing in front-focal areas and green fringing behind the focal plane. Most visible at wide apertures.
- LATERAL (TRANSVERSE) CA: Different wavelengths focus at the same distance but are magnified differently, causing colour fringing at image periphery. Always zero at the optical centre, increasing toward corners.

Visual Characteristics for Detection:

- Purple/cyan fringing along bright-to-dark edges: branches against sky, hair against bright backgrounds, window frames, metallic edges.
- Fringing direction: lateral CA fringing radiates outward from the image centre — always tangential to a circle centred on the optical axis.
- Aperture dependence: CA is most severe at wide apertures (f/1.4–f/2.8) and diminishes as the aperture is stopped down.
- Correction artifacts: Modern cameras and software correct CA, but overcorrection leaves a faint opposing-colour fringe.

Chromatic Aberration Index: The Abbe Number (V) = $(n_D - 1) / (n_F - n_C)$, where n_D , n_F , n_C are refractive indices at Fraunhofer D, F, C spectral lines. Higher V = lower dispersion = less CA.

■ NOTE: To add authentic CA in post-processing: split the image into RGB channels; scale the red channel outward by 0.3–0.8% and the blue channel inward by the same amount relative to centre. Merge. This simulates lateral CA precisely.

1.2 — VIGNETTING

■ ERRATA — Doc 2 § 1.2 FORMULA CORRECTION

The vignetting formula was corrupted in the original PDF. The correct physical equation is: $I(\theta) = I(0) \times \cos^4(\theta)$, where θ is the off-axis angle from the optical centre. This is the natural vignetting law (cos-fourth law) governing optical vignetting.

Vignetting is the reduction of image brightness or saturation at the periphery compared to the centre. It is produced by multiple physical mechanisms and is present to some degree in virtually all photographic lenses, particularly at maximum aperture.

Three Mechanisms:

- OPTICAL VIGNETTING (NATURAL): Caused by the lens barrel blocking oblique light rays. Intensity fall-off follows the $\cos^4(\theta)$ law. Produces smooth, centrally-symmetric darkening.
- MECHANICAL VIGNETTING: Caused by lens hoods, filter rings, and multi-element barrel walls clipping the light cone at image corners. Produces sharper, more geometric corner darkening.
- PIXEL VIGNETTING: Pixels at the image periphery are angled relative to incoming light and collect fewer photons, amplifying optical vignetting on digital systems.

Optical Vignetting Formula:

$I(\theta) = I(0) \times \cos^4(\theta)$ [where θ = angle from optical axis]

- Brightness: corners are 0.5–2 stops darker than centre at wide apertures; diminishes significantly when stopped down 2–3 stops.
- Shape: elliptical rather than circular in most modern lens designs.
- Colour shift: heavy vignetting often produces a slight colour shift in addition to luminance darkening.
- AI images lack organic vignetting — any vignetting present is applied as a uniform post-process filter rather than following the $\cos^4$ optical law.

1.3 — LENS DISTORTION

Lens distortion causes straight lines in the real world to appear curved in the photograph. It is a geometric aberration that does not affect image sharpness but changes the spatial mapping of the scene.

- BARREL DISTORTION: Image appears to bulge outward at the centre. Characteristic of wide-angle and zoom lenses at widest settings.
- PINCUSHION DISTORTION: Image appears to compress toward the centre. Characteristic of telephoto lenses and zoom lenses at longest settings.
- MUSTACHE (WAVE) DISTORTION: Barrel-type near the centre and pincushion-type at corners. Characteristic of contemporary ultra-wide lenses.
- Real architectural photography always shows some residual distortion; perfectly straight lines in a wide-angle interior shot indicate either AI generation or heavy software correction.

1.4 — LENS FLARE

Lens flare occurs when bright light sources enter the lens and scatter within the optical system, producing veiling glare and discrete flare artifacts.

- VEILING GLARE (HAZE FLARE): Diffuse reduction in contrast from scattered light. Manifests as milky, low-contrast appearance near the light source.
- POLYGON FLARE SPOTS: Discrete bright spots lying on a straight line passing through the optical centre. Shape corresponds to the lens aperture geometry.
- ANAMORPHIC STREAK FLARE: Characteristic horizontal streaks produced by anamorphic lens designs — blue or cyan horizontal lines from bright point sources.
- Real flare spots always lie on a straight line passing through the optical centre. Flare spots off this axis indicate AI generation.
- The polygon count of flare spots matches the number of aperture blades.

1.5 — DIFFRACTION

Diffraction occurs when light passes through a narrow aperture opening. At very small apertures (f/16–f/22 and beyond), diffraction limits resolving power, causing uniform image softening.

Airy Disk Diameter: $d = 2.44 \times \lambda \times f_number$ [where λ = wavelength of light in metres]

■ **NOTE:** At f/22 with green light (550nm), the Airy disk diameter is approximately 27.5 micrometres — larger than the photosite pitch of most modern sensors. Real architectural and landscape photography often shows slight diffraction softening from use of f/11–f/16 for maximum depth of field.

SECTION 2 — SENSOR AND FILM ARTIFACTS

■ ERRATA — Doc 2 § 2 CORRECTION	ORIGINAL TEXT treated sensor/film artifact absence as a primary indicator of AI generation. This is REVISED. Denoise processing, phone computational photography (multi-frame HDR, pixel binning), image resizing, and recompression can all suppress grain and noise in authentic photographs. Noise and grain are context cues; their absence may result from denoise or computational photography, not necessarily from AI generation. Always consider the capture pipeline before concluding.
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2.1 — ISO NOISE (DIGITAL SENSOR NOISE)

ISO noise arises from amplification of the electrical signal from the camera's image sensor. At low ISO (100–400), signal-to-noise ratio is high and noise is minimal. As ISO increases, more amplification is applied, amplifying both the photographic signal and background electronic noise.

Types of Digital Sensor Noise:

- SHOT NOISE (PHOTON SHOT NOISE): Random variation in photons collected per pixel, governed by Poisson statistics. Irreducible and present in all photography. Formula: $\sigma_{\text{shot}} = \sqrt{N_{\text{photons}}}$.
- READ NOISE: Electronic noise introduced when the sensor reads accumulated charge. A fixed noise floor — the limiting factor in shadow detail at low ISO.
- FIXED PATTERN NOISE (FPN): Systematic, repeatable noise patterns from pixel-to-pixel variation in sensitivity. Produces structured artifacts — horizontal or vertical banding.
- THERMAL (DARK CURRENT) NOISE: Electrons generated by thermal energy, independent of light exposure. Increases with sensor temperature and exposure duration. Significant in long-exposure astrophotography.

Spatial Characteristics of Digital Noise:

- Digital noise manifests as individual pixel-level variation — a granular pattern at the pixel scale of the sensor.
 - COLOUR NOISE (CHROMINANCE): Random colour variation in pixels, most visible in shadows and flat areas. Appears as multicoloured speckles.
 - LUMINANCE NOISE: Variation in pixel brightness without colour shift. Appears as monochromatic grain.
 - Noise is spatially uncorrelated — adjacent pixels have independent noise values.
 - Noise is channel-dependent: at high ISO, the blue channel typically shows more noise than red or green.
- *NOTE: To add authentic ISO noise in post-processing: use luminance noise at 60% and colour noise at 40% of total applied grain. Apply at pixel level, not as a blurred texture overlay. The noise must be spatially random — not patterned or repeating.*

2.2 — COMPUTATIONAL PHOTOGRAPHY ARTIFACTS (PHONE CAMERAS)

Modern smartphones use aggressive computational pipelines that fundamentally alter optical physics compared to traditional large-sensor DSLRs. These pipelines create authentic-looking artifacts that must not be misclassified as AI generation.

- **MULTI-FRAME NOISE REDUCTION SMEARING:** In low-light conditions, multi-frame stacking obliterates fine textures such as hair strands or fabric weaves, creating a 'watercolour' effect. This is a genuine mobile artifact — not an AI smoothing tell.
- **UNSHARP MASKING HALOS:** Mobile ISPs apply heavy edge sharpening, creating distinct white halos along high-contrast edges (dark building against bright sky). These mimic generative edge-bleeding but are mathematically distinct.
- **MULTI-FRAME HDR TONE-MAPPING:** Merging underexposed and overexposed frames heavily compresses midtones and often eliminates natural highlight clipping entirely, creating a flat, hyper-illuminated aesthetic that mimics AI lighting.
- **PORTRAIT MODE DEPTH ESTIMATION FAILURES:** Synthetic depth-of-field relies on depth-mapping software, frequently causing edge-detection failures where transparent objects or complex silhouettes are incorrectly blurred into the background.
- **PIXEL BINNING TEXTURE LOSS:** Low-light pixel binning combines adjacent photosites, trading resolution for sensitivity. This creates a specific low-resolution texture pattern distinct from AI generation.

2.3 — FILM GRAIN (SILVER HALIDE GRAIN)

Film grain is produced by the random distribution of silver halide crystals in the photographic emulsion. Unlike digital noise, film grain has characteristic clumping behaviour and interacts with image tone in a specific way.

Film Grain Physics:

- **Crystal Size and ISO:** Higher ISO (more sensitive) emulsions use larger silver halide crystals, producing visually coarser grain.
- **Tonal Response:** Film grain is most visible in mid-tone grey areas and least visible in pure blacks and pure whites.
- **Colour Film Grain Structure:** Three separate emulsion layers (cyan, magenta, yellow), each with its own grain structure, slightly spatially offset.

Characteristic Film Stock Profiles:

Film Stock	ISO	Grain Character	Colour Response	Classic Use
Kodak Portra 400	400	Fine, creamy, organic clumping	Warm skin tones, muted greens	Portrait, lifestyle
Kodak Ektar 100	100	Ultra-fine, almost invisible	Saturated, vivid colours	Landscape, commercial
Fujifilm Pro 400H	160	Very fine, neutral-cool	Natural, slightly cool	Fashion, portrait
Ilford HP5 Plus	400	Classic B&W; medium grain	B&W; wide exposure latitude	Street, reportage
Kodak T-MAX P3200	3200	Coarse, pronounced clumping	B&W; high contrast shadow	Low light, music

Film Stock	ISO	Grain Character	Colour Response	Classic Use
Cinestill 800T	800	Medium, with halation	Tungsten balance, cinematic	Night, artificial light

2.4 — MOTION BLUR

- SUBJECT MOTION BLUR: A moving subject against a stationary background produces a directional streak. Streak length is proportional to subject speed and inversely proportional to shutter speed.
- CAMERA SHAKE BLUR: Produced by hand-holding at slow shutter speeds. Affects the entire frame uniformly. Non-linear (camera rotation produces arc motion) and typically double-edged.
- ROLLING SHUTTER DISTORTION: CMOS sensors read the image from top to bottom over a finite time period. Fast lateral motion causes vertical subject elements to lean — the 'jello effect.'
- Real motion blur shows intensity tapering at the blur extremities. AI motion blur (when present) is often applied uniformly without intensity tapering and may not correspond to a physically plausible motion vector.

SECTION 3 — DEPTH OF FIELD AND BOKEH PHYSICS

3.1 — DEPTH OF FIELD PHYSICS

Depth of field (DOF) is the zone of acceptably sharp focus in an image. Elements within the DOF appear sharp; elements outside appear increasingly blurred.

$$DOF = 2 \times u^2 \times N \times c / f^2$$

[u=focus distance, N=f-number, c=circle of confusion, f=focal length]

DOF Variable	Effect	AI Detection Note
Aperture (f-number)	Dominant variable. f/1.4 = shallow; f/16 = deep	AI images frequently depict impossibly shallow DOF requiring apertures below f/0.5
Focal Length	Longer focal lengths produce shallower apparent DOF for the same framing	Background compression should match implied focal length
Subject Distance	DOF decreases dramatically at close focus distances	Macro photography has DOF measured in millimetres
Sensor Size	Larger sensors produce shallower DOF at same aperture and framing	Full-frame bokeh vs phone-camera bokeh should look different

3.2 — BOKEH CHARACTERISTICS

- APERTURE BLADE GEOMETRY: Out-of-focus point light sources render as the shape of the lens aperture. A 7-blade aperture produces heptagonal bokeh at intermediate apertures.
- CAT'S-EYE BOKEH: At image corners, circular bokeh becomes elliptical due to vignetting of the bokeh circle by the lens barrel. Real bokeh always shows this at image corners for wide-aperture lenses.

- ONION RING BOKEH: Caused by aspherical lens elements. The bokeh circles show concentric ring patterns.
- AI-generated background blur is typically applied as a uniform Gaussian blur without aperture polygon geometry. Real bokeh circles are rarely perfectly circular.

SECTION 4 — REFLECTION, REFRACTION, AND LIGHT PHYSICS

4.1 — DIFFUSE REFLECTION

Diffuse reflection (Lambertian reflection) occurs when light strikes a rough surface and is scattered in multiple directions regardless of the angle of incidence.

Lambert's Cosine Law: $I_{diffuse} = I_{source} \times k_d \times \cos(\theta)$ [θ = angle between light direction and surface normal, k_d = diffuse coefficient]

Diffuse Reflection Characteristics:

- No specular component — the surface produces no mirror-like highlights.
- Even brightness from all viewing angles (Lambertian behaviour).
- Colour of reflected light = surface colour x light colour (multiplicative).
- INTER-REFLECTIONS: Diffuse surfaces bounce coloured light onto adjacent surfaces — a red wall will bleed red light onto a white ceiling, creating a subtle pink cast. AI images often miss this inter-reflection colouring.

4.2 — SPECULAR HIGHLIGHTS

Specular reflection occurs when light bounces off a smooth surface at the mirror angle. It produces bright, localised highlights revealing the shape and position of light sources.

Phong Model: $I_{spec} = I_{source} \times k_s \times (R \cdot V)^n$ [R =reflection vector, V =view vector, n =shininess exponent]

Material	Specular Type	Highlight Shape	Colour of Highlight
Mirror / Chrome	Perfect specular	Point or geometric shape of light source	White (light source colour)
Wet skin / Sweat	Moderate specular	Soft, somewhat defined	White (slightly warm)
Silk / Satin fabric	Anisotropic specular	Elongated along weave direction	Near-white
Human eye (cornea)	High specular	Sharp catchlights from light sources	White (light source colour)
Matte skin	Very low specular	Barely visible, broad	Skin colour + white blend
Glossy plastic	High specular	Defined but not perfectly sharp	White (light source colour)

4.3 — THE INVERSE-SQUARE LAW OF LIGHT

The Inverse-Square Law is the fundamental physical law governing how light intensity diminishes with distance from a point source. It is one of the most frequently violated laws in AI-generated imagery.

$$I = P / (4 \times \pi \times d^2)$$

[I=intensity, P=power of source, d=distance from source]

Equivalently: doubling the distance from a light source reduces its intensity to one-quarter of the original value.

- A subject 2 metres from a flash receives 4x more light than a subject 4 metres away.
- Backgrounds in portrait photography receive significantly less flash illumination than foreground subjects — creating characteristic dark backgrounds in off-camera flash photography.
- AI images frequently show uniform ambient illumination that does not follow the inverse-square fall-off.
- The inverse-square law applies to point sources. Large, diffuse sources (overcast sky, large softboxes) produce more gradual fall-off.

Illumination Ratio: $E1/E2 = (d2/d1)^2$

[E=illuminance, d=distance from source]

4.4 — FRESNEL EFFECT AND GRAZING ANGLE REFLECTIVITY

The Fresnel equations describe how the reflectivity of a surface changes with the angle of incidence. At near-perpendicular incidence, most non-metallic surfaces absorb most light and reflect little. At grazing angles (near-parallel to the surface), reflectivity increases dramatically.

- This produces the characteristic wet-road or polished-floor effect where the floor appears highly reflective at the far end of a corridor (low angle from that vantage point) but not at close range.
- Real human skin shows increased specularity at the edges of the face (grazing angle) compared to the frontal plane.
- AI models frequently miss Fresnel behaviour — flat surfaces are either uniformly reflective or uniformly diffuse without angle-dependent variation.

SECTION 5 — EXPOSURE, TONAL RESPONSE, AND COLOUR PHYSICS

5.1 — THE EXPOSURE TRIANGLE

Parameter	Effect on Exposure	Secondary Effect
Aperture (f-number)	Larger aperture (lower f-number) = more light	Shallower depth of field; lens aberrations more pronounced
Shutter Speed	Longer exposure = more light	Motion blur on moving subjects; camera shake at slow speeds
ISO Sensitivity	Higher ISO = more signal amplification	Increased sensor noise; reduced dynamic range

5.2 — DYNAMIC RANGE AND HIGHLIGHT/SHADOW CLIPPING

■ **ERRATA — Doc 2**
§ **5.2 CORRECTION**

ORIGINAL TEXT stated: 'AI images rarely show true clipping.' This is **REVISED**. Contemporary diffusion models trained on HDR content routinely simulate highlight clipping. Meanwhile, multi-frame HDR pipelines in modern smartphones actively suppress clipping in authentic photographs. This heuristic is unreliable in isolation. Do not use clipping presence or absence as primary AI evidence.

- **HIGHLIGHT CLIPPING:** Blown-out windows, sun discs, specular highlights. Real photographs frequently clip highlights in HDR scenes.
- **SHADOW BLOCKING:** Underexposed areas lose detail completely. Real photography in mixed lighting routinely shows blocked shadows.
- **HDR TONE-MAPPING ARTIFACT:** When digital HDR processing is applied, it produces characteristic halo artifacts around high-contrast edges.

5.3 — WHITE BALANCE AND COLOUR TEMPERATURE

Light sources have characteristic colour temperatures measured in Kelvin (K). Camera white balance adjusts the image to neutralise colour cast, but real photographs frequently contain residual colour casts or mixed lighting that white balance cannot fully correct.

Light Source	Colour Temp (K)	Colour Cast if Unbalanced
Candlelight	1,800–2,000 K	Very warm orange-red
Tungsten/Incandescent	2,800–3,200 K	Warm orange-yellow
Halogen	3,000–3,500 K	Warm yellow
Fluorescent (cool)	4,000–5,000 K	Green-cyan cast
Daylight (noon)	5,500–6,000 K	Neutral
Overcast sky	6,500–7,500 K	Cool blue
Open shade / Blue sky	8,000–10,000 K	Strong blue-cyan
LED (varies)	2,700–6,500 K	Variable — often greenish

Mixed lighting is a powerful authenticity indicator. Real indoor photography often shows warm tungsten light on a subject with cool daylight coming through a window. AI images frequently apply a uniform, neutralised colour temperature to the entire scene.